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LGCA-Net: A time-series anomaly detection method for multiple types of agricultural sensors

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ABSTRACT

Agricultural sensors play a crucial role in guiding agricultural production. However, the quality of data collected by these sensors is often compromised by the complex environmental conditions of farmlands and the diversity of crop management practices. Particularly in precision agriculture, even minor anomalies in data can lead to significant deviations in critical decisions, such as estimation of crop water demand, fertilization scheduling, and pest or disease management strategies, thereby resulting in substantial losses. To address this challenge, this study introduces a universal time-series anomaly detection method for multi-type agricultural sensors employing the fusion of global and local features, termed LGCA-Net. To capture the short-term change rule of data, a multiscale convolutional neural network (MCNN) is employed to extract detailed local information. For identifying the long-term change rule of data, a transformer neural network is employed to capture the global correlation within the data. Finally, the fusion of global and local information is achieved via a cross-attention mechanism, which enhances the anomaly recognition ability of the model under different agricultural sensor data. To evaluate the effectiveness of the proposed model, experiments were conducted using two agricultural scenario datasets from greenhouses and fields. The results demonstrate that the proposed method achieves F1-scores of 0.9580 (field) and 0.9721 (greenhouse), which comprehensively measures precision and recall. Compared with the single-structure model, the Accuracy (ACC) of abnormal sample recognition increased by an average of 6.65% (field) and 6.67% (greenhouse). In summary, the proposed LGCA-Net model adapts to the dual characteristics of short-term fluctuations and long-term cycles in agricultural time series through the parallel fusion of global and local features. It overcomes the limitations of single-structure models in feature capture and provides a reliable anomaly detection solution for multi-type of agricultural sensors.

1. Introduction

Agricultural sensors constitute an important component of intelligent agriculture and perform critical functions during the digital transformation of agriculture, including crop growth monitoring, soil moisture measurement, and meteorological environment observation [1]. Data collected from these sensors provide a scientific foundation for downstream processes such as irrigation and fertilization scheduling, pest and disease early warning systems, and yield prediction. However, under real agricultural conditions, sensors are highly susceptible to multiple external influences, such as strong winds, heavy rainfall, farming, and sensor aging, which introduce noise and outliers in the collected data. These abnormal data directly degrade the accuracy of subsequent analysis, modeling, and decision-making, thereby significantly restricting the applicability and operational efficiency of intelligent agriculture.

Common agricultural monitoring indicators include air temperature, relative humidity, and soil moisture content. In agricultural production systems, these indicators are not independent but are simultaneously influenced by climate change (seasonal temperature difference, extreme

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weather, rainfall spatial and temporal distribution), crop growth stage (sowing, seedling stage, jointing, grouting, and maturity), agricultural management measures (irrigation, fertilization, farming, and pest control), and other multiple effects. These factors produce highly complex nonlinear relationships. In addition, agricultural monitoring differs substantially from industrial settings and exhibit strong nonstandard characteristics, such as diverse sensor layout conditions, uncontrollable environmental disturbances, difficulty in maintaining long-term consistency of measurement accuracy, and limited comparability across different fields.

In the context of multifactor interweaving and pronounced dynamic changes, conventional anomaly detection methods, such as the threshold method [2,3] and moving average method [4], exhibit inherent limitations in agricultural applications. Threshold-based methods significantly depend on expert-defined judgment criteria [5], which limits adaptability across regions, crops, and highly variable climatic conditions. Although moving average methods have achieved satisfactory performance in specific settings, they result in loss of detailed information and reduced recognition accuracy in complex environments [6]. Furthermore, agricultural production is influenced by multiple factors, including seasonal changes, climate fluctuations, and crop growth cycles, resulting in dynamic changes in water usage patterns and data characteristics [7]. Once established, traditional anomaly detection models exhibit limited parameter flexibility and weak adaptability to newly emerging data patterns. This significantly restricts the intelligent and refined development of agricultural water resource management.

With the continuous development of artificial intelligence technology, machine learning methods have attracted increasing attention in anomaly detection owing to their powerful feature-learning and pattern-recognition capabilities. In 2021, Zamry et al. [8] proposed a lightweight anomaly detection scheme combining incremental principal component analysis (CCIPCA) and a one-class support vector machine (OCSVM). The experimental results show that this scheme achieves excellent performance in detection accuracy, memory efficiency, computational complexity, and communication overhead. In 2023, Utkin et al. [9] proposed an attention-based isolation forest (ABIForest) for anomaly detection. This method enhances the anomaly detection performance by incorporating the Nadaraya–Watson attention mechanism. The experimental results demonstrated that the ABIForest outperformed the traditional isolation forest (iForest) on multiple synthetic and real-world datasets. In 2024, Chua et al. [10] explored the use of an isolation forest algorithm for detecting network traffic anomalies. Their study indicated that isolation forest exhibited outstanding performance in detecting network traffic anomalies, with an accuracy rate exceeding 90%. Zhang et al. [11] proposed a method for electric vehicle battery fault prediction and diagnosis based on an improved gradient boosting decision tree (GBDT)–iForest)–boxplot. By jointly considering the battery characteristics and driving behaviors, accurate and effective anomaly detection was achieved. In 2025, Verma et al. [12] proposed an anomaly detection method combining optimal partitioning (OP) and extreme learning machine (ELM), demonstrating strong performance on highly imbalanced datasets. Despite significant progress in machine learning-based methods for anomaly detection, their applications remains predominantly concentrated in industrial sectors, and their implementation in agriculture continues to encounter considerable challenges. Industrial environments are relatively stable and highly controllable, with relatively uniform fault patterns. Therefore, traditional machine learning algorithms based on specific assumptions on data distribution and features often demonstrate optimal performance. However, data from agricultural environments are highly susceptible to factors such as weather conditions and crop growth stages, exhibiting a range of normal fluctuations and frequent “normal yet extreme” patterns. When real-world data deviate from predefined algorithmic assumptions, traditional machine learning models struggle to accurately capture the distribution patterns of the data, leading to a significant decline in their detection performance.

Deep learning, as a branch of machine learning, can automatically learn complex patterns and feature representations from massive datasets leveraging multilayer neural networks [13]. Owing to its superior feature-learning capability compared with traditional machine learning, it has emerged as a major research focus in anomaly detection. Deep learning-based anomaly detection methods are primarily categorized into two types. The first is the method of reconstruction by learning the patterns of normal data. These include reconstruction models based on autoencoders such as deep autoencoder [14], long short-term memory (LSTM)-capsule autoencoder [15], autoencoder based on neural network [16], multi-channel multi-scale convolution attention variational autoencoder (MCA-VAE) [17], and variational autoencoders based on association discrepancy [18], as well as generative adversarial networks (GAN) such as transformer-based GAN for anomaly detection of time series data (TGAN-AD) [19], LSTM-Based VAE-GAN [20], and federated time series anomaly detection method based on the Sobolev–Wasserstein GAN (FedSw-TSAD) [21]. Autoencoder-based methods identify anomalies by evaluating the degree of reconstruction, whereas GAN-based methods rely on discriminator outputs for anomaly judgment. The second category emphasizes real-time anomaly discrimination of future data based on the change patterns of historical data, among which LSTM is a commonly used model owing to its structural suitability for time-series feature extraction. Numerous studies have demonstrated the effectiveness of LSTM-based models in time-series anomaly detection, including Gopali [22], Noh [23], Lee [24], and Gomolka [25]. Additionally, some models based on convolutional neural network (CNN), transformers, and their fusion variants have demonstrated excellent performance in time-series anomaly detection. Examples include LSTM-CNN [26], CNN-LSTM fused with ARMA [27], and TCF-Trans [28].

Although deep learning methods eliminate the need for manual feature design, they can automatically capture the complex latent structures of data and have demonstrated robust performance across diverse domains. Agricultural data are closely linked to climate change and crop growth, resulting in strong temporal characteristics. However, anomaly detection methods based on data reconstruction often fail to fully capture the inductive bias of adjacent points at each time point in time-series data. This inductive bias indicates that, for each time point in a time series, its immediate neighbors are more relevant or influential than distant points [29]. For instance, traditional autoencoders may lose the local correlation information of adjacent points owing to compression of the encoding dimension, making it challenging to capture detailed features. In addition, agricultural production requires real-time and dynamic monitoring. Reconstruction-based methods require the calculation of the entire sequence for each detection, consume a large amount of resources, and make real-time detection challenging.

Therefore, this study focuses on an anomaly-detection method based on real-time discrimination. This method can effectively extract the contextual information of time-series data while emphasizing the influence of neighboring points, thereby rendering it more suitable for real-time anomaly detection of time-series data. However, different models exhibit varying feature-capturing capabilities. The agricultural data exhibits significant differences in their variation patterns across different timescales. On a short-term scale, sudden temperature fluctuations, unexpected rainfall, and irrigation can lead to rapid changes in data such as soil moisture. Substantial differences exist in the scales of the fluctuations, and the features interact with each other within the feature space. Over long time scales, agricultural data follow pronounced seasonal and periodic patterns and are also influenced by long-term factors such as climate change and adjustments in the planting structure. Although data from the same crop growth cycle showed similar trends, factors such as climate can cause an overall shift in long-term data.

Most existing real-time discriminative anomaly-detection methods are based on a single model or serial fusion. A single model struggles to

adapt to the highly variable characteristics of agricultural environments. Serial fusion method, exemplified by the CNN-LSTM model, perform feature extraction sequentially, with the CNN output serving as the sole input to LSTM. The learning performance of LSTM is completely dependent on the quality of feature extraction by the CNN. If the CNN fails to capture key information, the subsequent learning of the LSTM deteriorates, and LSTM cannot compensate for the shortcomings of the previous module via self-adjustment.

To address these limitations, this study proposes a time-series anomaly detection method (LGCA-Net) for multiple types of agricultural sensors based on the parallel fusion of global and local features. It primarily comprises the following three components:

(1) To address the short-term variation patterns of data, a local feature extraction module based on a multi-scale convolutional neural network is constructed. This module utilizes convolution kernels of different sizes to mine local feature information from the data, enabling the capture of minor fluctuations and instantaneous changes in the data.

(2) To address the long-term variation patterns of data, a global feature extraction module based on Transformer is built. Through the multi-head self-attention mechanism, this module calculates the attention distribution of multiple subspaces in parallel, capturing global correlations of the data from multiple perspectives. This allows the module to identify significant differences in long-term changes, such as those caused by different seasons, climates, and crop growth cycles.

(3) A global and local feature fusion module based on the cross-attention mechanism is constructed. The cross-attention mechanism establishes connections between different features, which facilitates the exchange and integration of information, forming a complete feature extraction system that spans from local to global, thereby enhancing the model's representational capacity for complex data.

In Section 2, we present information pertaining to the algorithm, including a general flow chart and the theory behind each component. In Section 3, we present the experimental data and model parameter configurations and verify our method's effectiveness through comparative experiments. In Section 4, we conclude this paper, analyze the limitations of the current study and provide an outlook on future directions of research.

2. Methodology

In this section, the overall algorithm workflow proposed in this study is briefly described. We then elaborate on the algorithmic principles of the three modules: multiscale local feature extraction, global feature extraction, and cross-attention mechanism-based feature fusion. Finally, the anomaly determination criteria and evaluation metrics employed in this study are introduced.

The algorithm proposed in this study is illustrated in Fig. 1. In the multiscale local feature extraction stage, three convolutional residual blocks of different scales are adopted for feature extraction. The extracted multiscale features are concatenated to form a multiscale local feature map, thereby providing fine-grained feature support for subsequent anomaly identification. The encoder structure of the Transformer was employed in the global feature extraction stage. Using the multi-head attention mechanism, the attention distribution of multiple subspaces is calculated in parallel, and the dependency relationships across different time spans are captured. This enables the construction of global features covering a long time series, preventing misjudgment caused by local fluctuations and the consequent neglect of abnormal overall trends. In the cross-attention-based feature-fusion stage, the self-attention mechanism is first applied to the global and local feature maps to obtain their respective query (Q), key (K), and value (V) matrices. Subsequently, the Q matrix of the global features is used to query the K and V matrices of the local features, whereas the Q matrix of the local features is used to query the K and V matrices of the global features. This process enables information interaction between the global and local features, thereby enhancing the ability of the model to learn from complex data. Finally, the interacting features are fused and input into the fully connected layer to generate the final anomaly detection result. This workflow can specifically capture data characteristics in anomaly detection for typical agricultural monitoring indicators (such as soil moisture content, temperature, and humidity), thereby improving identification accuracy.

2.1. Multi-scale local feature extraction

The local feature extraction module employed a one-dimensional (1D) CNN. The 1D-CNN shares similar principles with the two-dimensional (2D) CNN, with the primary distinction that the input data comprises one-dimensional time-series data. Convolution kernels slide over the input sequence to compute the convolution results at each position, forming a new feature map wherein each element represents the feature of a local region in the input sequence [30].

However, in agricultural applications, temporal data patterns often exhibit the characteristics of coexisting multiscale features, such as short-term precipitation fluctuations and medium-term changes in growth status. Conventional 1D-CNNs employ a single convolution

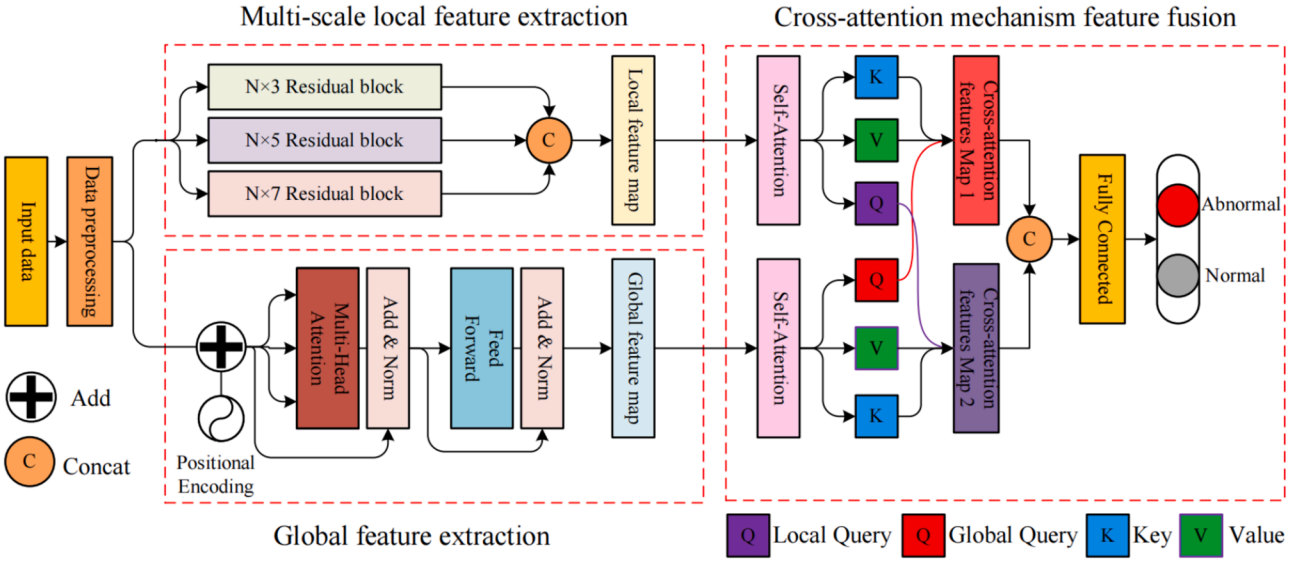


Fig. 1. Overall algorithm flowchart of LGCA-Net for agricultural sensor time-series anomaly detection.

kernel with a fixed receptive field, which makes it challenging to capture information at different time granularities.

Therefore, this study employed a multiscale convolutional structure. By integrating multi-scale techniques with a 1D-CNN, it leverages the high-efficiency of CNN in extracting local features and also incorporates multilevel patterns in agricultural data via multi-scale design. Examples of these patterns are sudden changes in daily temperature and humidity and the stability of the growth environment within a week. The specific structures are shown in Fig. 2.

where kernel sizes 1–3 correspond to convolution kernels with different scales (covering short, medium, and long time spans, respectively). By sliding these multi-scale convolution kernels along the temporal dimension of the agricultural sensor time-series data, local features corresponding to different time spans in the sequence are extracted, which provides diverse fine-grained feature representations for subsequent global feature fusion.

In addition, to mitigate the potential loss of temporal information caused by deep-layer convolutions, this study constructed convolutional residual blocks to replace single-layer convolutions. In the convolutional residual block, the original input information can be transmitted to the subsequent layers without loss through skip connections. This feature is crucial in the context of agricultural time-series data. Sudden short-term meteorological changes (such as heavy rains) may exert long-term influences on crop growth. The residual block ensures that key information is not excessively smoothed by the convolution operation. The specific structures are shown in Fig. 3.

In this study, two skip connection methods are employed. When the input and output dimensions are identical, residual block (a) uses identity mapping to realize the skip connection, which prevents unnecessary computation and ensures the integrity of the original features. When the input and output dimensions are differ, the residual block (b) performs a dimension alignment through a 1×1 1D convolutional layer to align the feature dimensions from the residual mapping path, thereby ensuring the feasibility of element-wise addition. In Fig. 3, N represents the number of output channels in the convolutional layer, which determine the number of output feature maps in the convolutional layer. Each channel can capture different types of feature information from the input data. K denotes the size of the convolutional kernel, which can control the temporal size of the receptive field of the convolution operation, thereby enabling the capture of local feature information at different scales. The BN is a batch normalization layer that can effectively alleviate the "internal covariate shift" and prevent gradient vanishing during later stages of training.

2.2. Global feature extraction

In the global feature extraction module, this study adopts the encoder part of the Transformer. Its structure comprises positional encoding, a multi-head self-attention mechanism, and a feedforward neural network. The self-attention mechanism can directly model the dependency relationships between any positions in a sequence. Unlike the RNN/LSTM, it does not need to transmit information point-by-point [31], making it particularly suitable for capturing global patterns in agricultural data. For example, it can simultaneously focus on the long-term correlation between "high temperature in summer" and "low temperature in winter," while traditional time-series models struggle to learn such long-distance dependencies owing to the vanishing gradient problem. The formula is as follows:

$$\text{Attention}\ (\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \mathit{softmax}\left(\frac{\mathrm{QK}^{\mathrm{T}}}{\sqrt{\mathrm{d_K}}}\right) \times V \tag{1}$$

where $Q, K,$ and V denote the query, key, and value vectors, respectively. Furthermore, softmax is the normalization function, $\mathrm{d_K}$ is the dimension of the vector, and $\mathrm{QK^T}$ is the similarity score between Q and K. The self-attention mechanism enables the model to perform weighted aggregation by calculating the relationships between the positions in the input sequence. Each self-attention head acquires information from different positions within a sequence. Subsequently, the multi-head attention concatenates the output matrices and passes them into a linear layer to obtain the final output of the multi-head attention:

$$\mathrm{head_i} = \ \text{Attention}\left(\mathrm{QW_i^Q}, \mathrm{KW_i^K}, \mathrm{VW_i^V}\right) \tag{2}$$

$$\text{MultiHead}\ (Q, K, V) = \text{Concat}\ (\ \mathrm{head_1}, \cdots, \ \mathrm{head_h})W^o \tag{3}$$

where "*Concat*" represents the concatenation operation, $head_i$ is the i-th attention head, and h is the number of attention heads. $\mathrm{W^o}$ denotes the matrix weight used for linear transformation of the concatenated matrix. $\mathrm{QW_i^Q}$, $\mathrm{KW_i^K}$, and $\mathrm{VW_i^V}$ are the linear transformation matrices of the i-th attention head. After the aforementioned processing, the long-range dependencies between features were obtained. Subsequently, the output of each attention layer is passed through a feed-forward neural network. The feed-forward neural network comprises two linear layers and a non-linear activation function called ReLU. For the input feature vector $\mathrm{h_t}$ at the t-th time step, the feedforward neural network is calculated as follows:

$$\mathrm{a_t} = \mathit{ReLU}(\mathrm{h_t W_{FFN1}} + \mathrm{b_{FFN1}}) \tag{4}$$

$$\mathrm{o_t} = \mathrm{a_t W_{FFN2}} + \mathrm{b_{FFN2}} \tag{5}$$

where $\mathrm{a_t}$ is the intermediate activation value; $\mathrm{o_t}$ is the final output. Furthermore, $\mathrm{W_{FFN1}}$ and $\mathrm{W_{FFN2}}$ are learnable weight matrices, and $\mathrm{b_{FFN1}}$ and $\mathrm{b_{FFN2}}$ are bias vectors. Finally, the output of the global feature map was obtained after normalization, residual connections, and global pooling.

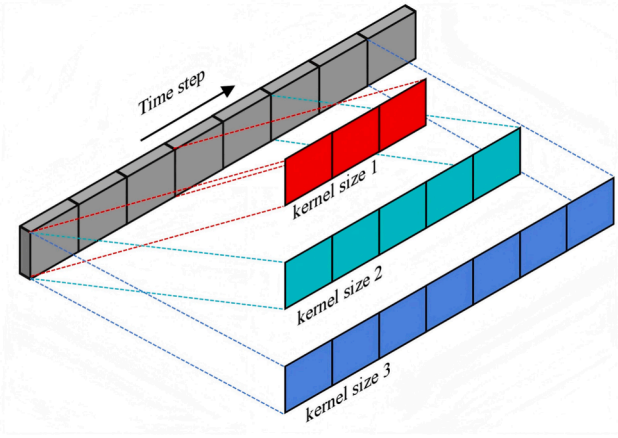


Fig. 2. Multi-scale local feature extraction diagram of MCNN.

2.3. Cross-Attention mechanism feature fusion

The Cross-Attention Mechanism is an important variant of the attention mechanism. This is in contrast to self-attention, which focuses only on dependency relationships within a single input sequence. Cross-attention addresses this limitation by enabling information exchange between two different sequences (commonly referred to as the "query sequence" and the "key-value sequence"). Its core idea is to allow one input sequence to "attend to" the relevant information in another sequence, thereby achieving effective integration across information sources. The specific process is illustrated in Fig. 4.

In Fig. 4, the query (Q) matrix is derived from the representation of the target sequence (Feature map 2), where the key (K) and value (V) are obtained from the representation of the source sequence (Feature map 1), A is the cross-attention weight, and the output feature map Z is generated by multiplying A with V.

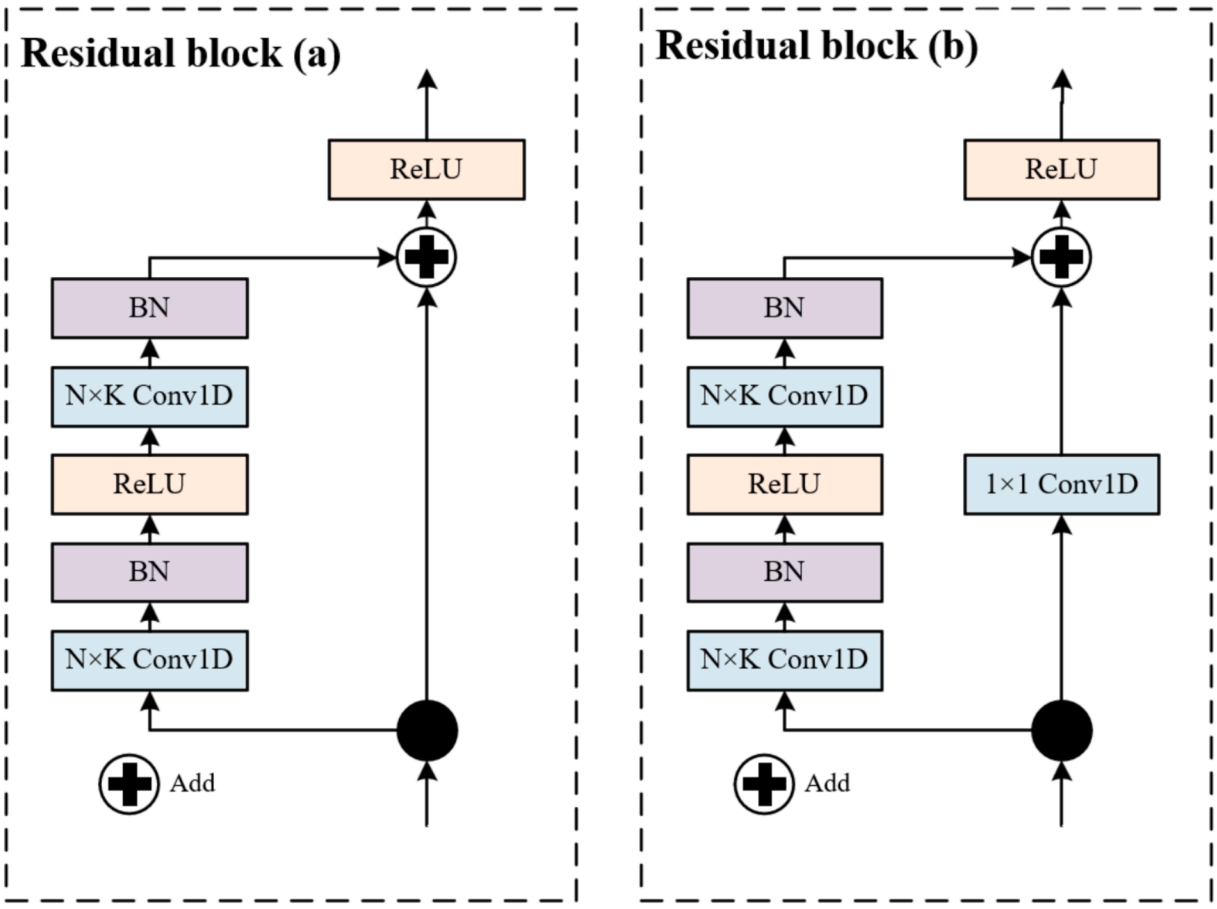


Fig. 3. Two residual block structures diagram for MCNN multi-scale feature extraction: (a) identity-mapped residual block; (b) 1 × 1 convolution-mapped residual block.

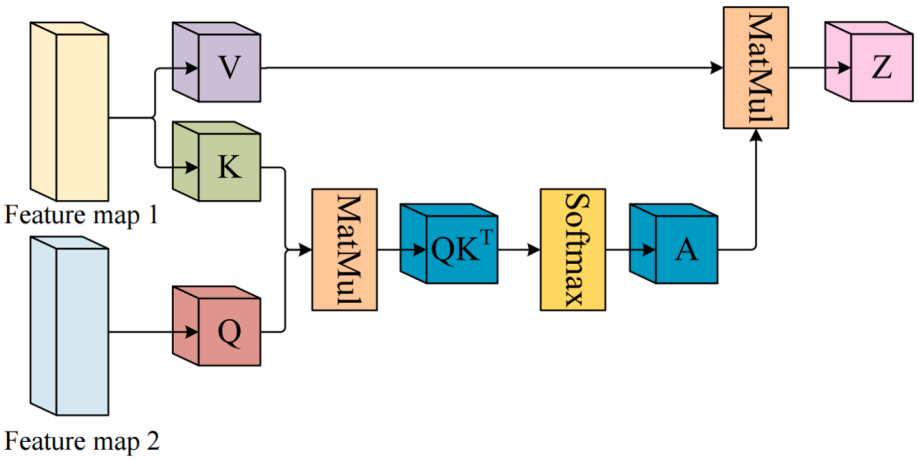


Fig. 4. Cross-attention structural diagram.

To fully preserve the interaction information between the global and local features, this study adopted a dual cross-attention mechanism. Dual cross-attention enables two-way interaction between "global → local" and "local → global," resulting in the following two core new capabilities: the "global → local" interaction calibrates local features from a global perspective, preventing misjudgment of data mutations caused by sudden environmental influences as anomalies (e.g., a rapid increase in soil moisture content owing to rainfall); the "local → global" interaction supplements global features with local details, preventing anomaly misjudgment resulting from the loss of key short-term signals during global modeling (e.g., short-term frequent irrigation leading to fluctuations in soil moisture content). After layer normalization and 1 × 1 convolution for dimension alignment, the key-value (KV) matrices of the global and local features can be dynamically adjusted according to the data characteristics to achieve multiscale adaptation. The specific formulae are as follows:

$$A^{(1)} = softmax\left(\frac{Q_{\text{local}}\, K_{\text{global}}^{\top}}{\sqrt{d_k}}\right) \tag{6}$$

$$Z^{(1)} = A^{(1)} V_{\text{global}} \tag{7}$$

$$A^{(2)} = softmax\left(\frac{Q_{\text{global}}\, K_{\text{local}}^{\top}}{\sqrt{d_k}}\right) \tag{8}$$

$$Z^{(2)} = A^{(2)} V_{\text{local}} \tag{9}$$

where d_k is the vector dimension and *softmax* is the normalization function. In first-layer cross-attention, Q of the local feature sequence is used as the query, and KV of the global features is used as the key value, thereby enabling global information to guide the local features. In second-layer cross-attention, the Q of the global feature sequence is used as the query, and the KV of the local features is used as the key value to achieve the enhancement of global features by local information. The resulting feature maps, denoted as $Z^{(1)}$ and $Z^{(2)}$, correspond to the local and global features, respectively, after cross attention. This "dual" cross-attention mechanism fully exploits feature interactions at different scales, incorporating both global information and strengthening local details. Finally, the two interacting features are concatenated, fused, and input into the fully connected layer to complete the detection and identification of abnormal data.

2.4. Criteria for abnormality determination and evaluation indicators

Given the inductive bias discussed earlier that the immediate neighboring points in a time series exert stronger influence than distant points, this study employed a sliding-window anomaly determination method. For each target point position, this method simultaneously inputs both the data within the previous window and the target data, and then outputs in real time whether the detected data are abnormal. This process is illustrated in Fig. 5.

To ensure full inclusion of the key reference information required for anomaly judgment while excluding redundant data that may interfere with the judgment and increase computational efficiency, this study employed the autocorrelation analysis method to determine the window length. This method quantifies the correlation between the data and its lagged versions, reveals the inherent temporal structure of the data, and ensures that the selected window size contains sufficient historical information to determine the anomaly of the current data point. The specific formula is as follows:

$$\rho_k = \frac{\sum_{t=k+1}^{n}(x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^{n}(x_t - \bar{x})^2} \tag{10}$$

where ρ_k represents the autocorrelation coefficient at a lag of k time steps, n denotes the length of the time series, and $\bar{x}$ denotes the mean of the time series. The steps to determine the window size based on the autocorrelation analysis are as follows:

1. Calculate the autocorrelation coefficients: For the given time series data, compute ρ_k for different lag orders k.
2. Identify the cutoff point: Find the lag order k at which the autocorrelation coefficient first drops below a specific threshold (set to 0.5 in this study). This value indicates that the correlation between the original data and its lagged versions is no longer significant.
3. Determine the window size: Set the window size to $k + 1$. This ensures that the window includes all historical data points with significant correlation.

To accurately evaluate the anomaly detection performance of the model proposed in this study, we consider that the proportion of abnormal samples is typically extremely low in practical scenarios. Consequently, the model tends to identify normal samples more readily. However, the cases of missed detection of anomalies and misjudgment of normal samples are precisely what we need to focus on, as these can directly impact the practicality of the method. Therefore, in this paper, evaluation metrics are selected based on the model's ability to recognize abnormal samples and the misjudgment of normal samples. Samples are classified into four categories according to the true labels and detection results: true positive (TP), false negative (FN), false positive (FP), and true negative (TN). Based on these four types of discrimination results, we selected the following four evaluation metrics to construct a model performance evaluation system:

1) Recall: Calculated as the ratio of the number of true positives to the total number of abnormal samples. The recall directly reflects the ability of a model to capture abnormal samples. A high recall rate indicates significantly few cases of missed detection of anomalies, which is a core requirement in anomaly detection tasks to ensure that potential anomalies are not overlooked. The calculation formula is as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{11}$$

2) Precision: It is measured as the proportion of the number of true positives in the total number of samples judged as abnormal. Precision focuses on reducing misjudgments of normal samples. The level of precision directly reflects the model's ability to distinguish normal fluctuations during the adaptive adjustment process, avoiding ineffective warnings caused by over-sensitivity. The calculation formula is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{12}$$

3) F1-Score: As the harmonic mean of recall and precision, it balances the potential conflict between the two, and comprehensively evaluates the overall performance of the method in terms of "minimizing missed detections" and "minimizing false alarms," particularly suitable for scenarios with imbalanced samples. The calculation formula is as follows:

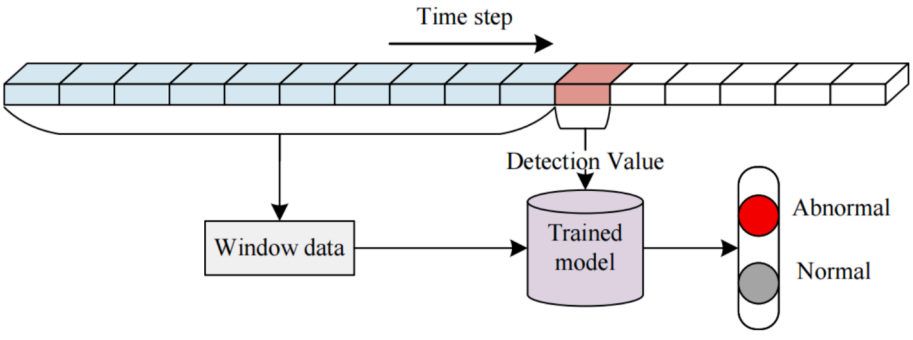


Fig. 5. Real-time anomaly detection flowchart based on sliding window.

$$F1 - Score = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

4) False Positive Rate (FPR): This was calculated as the ratio of the number of false positives to the total number of normal samples. The FPR quantifies the proportion of normal samples that are incorrectly predicted as abnormal. The FPR reflects the ability of the anomaly-detection model to correctly identify normal samples. A low FPR indicates that the model can prevent misclassifying normal samples more accurately, thereby reducing ineffective interference. By contrast, a high FPR may render the model useless owing to excessive false alarms. The calculation formula is as follows:

$$FPR = \frac{FP}{TN + FP} \tag{14}$$

3. Experimental results and discussion

3.1. Data Introduction and Preprocessing

This study conducted experimental analyses on common monitoring indicators in two agricultural scenarios: fields and greenhouses. For the field scenario, soil moisture content data in the 0–20 cm and 20–40 cm soil layers were selected. Soil moisture content is a fundamental environmental parameter in dryland and irrigated agriculture [32]. Anomaly detection in time-series data is critical for ensuring crop growth and improving water use efficiency. In the greenhouse scenario, air temperature and humidity data were selected. Time-series characteristics and anomaly patterns of air temperature and humidity are typical in various controlled agricultural environments as core indicators for environmental monitoring in protected agriculture [33]. Collectively, they constitute the key time-series monitoring data for two typical scenarios characterized by natural and controlled conditions in agricultural production, thereby providing strong representativeness. The corresponding monitoring environments are shown in Fig. 6.

To evaluate the generalization of the model under different time spans and data frequencies, the field and greenhouse data were collected over distinct time periods and sampling intervals. The specific data parameters are listed in Table 1.

After expert annotation, the field dataset contained 33,650 records, of which 32,885 were normal and 765 were abnormal. The greenhouse dataset contained 10,584 records, with 10,524 normal and 60 abnormal data. To accurately evaluate the anomaly detection performance of the model, we adopted a stratified random splitting method, splitting each stratum of the dataset independently at an 8:2 ratio before recombining them, so as to ensure that the test samples cover all annotated anomaly cases in this study. As reflected in the data distribution, the samples were imbalanced, and the proportion of abnormal data was relatively low. If the model is trained directly, overfitting may occur, and the model will tend toward the majority class (normal data), resulting in suboptimal detection of abnormal data. Therefore, to ensure effective detection performance of the model, this study performed a synthetic minority oversampling technique (SMOTE) oversampling operation on the training set data. By calculating the distances between abnormal samples, synthetic samples were generated between similar samples instead of simple replications. This ensured that overfitting is avoided with a balanced training sample.

Table 1
Description of field and greenhouse experimental data parameters.

Scenario	Input Elements	Time Range	Sampling Interval
Field	0–20 cm soil moisture content 20–40 cm soil moisture content	2021/1/1–2022/12/31	1 h
Greenhouse	Air temperature Air relative humidity	2024/10/1–2024/10/15	5 min

In addition, in the anomaly detection task, two key issues require attention: the false negative of anomalies (missed detection) and the false positive of normal samples (misjudgment as anomalies). Distinguishing abnormal samples that closely resemble normal samples or normal samples that resemble abnormal samples remains significantly challenging. Although SMOTE achieves "quantity balance," the model's attention to the minority class remains "passive." To enable the model to "focus actively" on these hard-to-distinguish samples, this study employs the Focal Loss function. Focal loss reduces the weight of easily distinguishable samples and increases the weight loss of hard-to-distinguish samples by introducing a modulating factor and weight factor [34]. Consequently, it effectively enhanced the anomaly detection performance of the model. The formula is as follows:

$$FocalLoss(p_t) = -\alpha_t(1 - p_t)^{\gamma}\log(p_t) \tag{15}$$

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{otherwise} \end{cases} \tag{16}$$

where p_t denotes the predicted probability of the model for the "target class." Furthermore, γ is the modulating factor, which is used to reduce the loss weight of easily-distinguishable samples and α_t is the weight factor, which is used to adjust the weights of positive and



(a) Field (b) Greenhouse

Fig. 6. Agricultural sensor deployment in two scenarios: (a) Field soil monitoring; (b) Greenhouse environmental monitoring.

negative classes. Typically, the weight of the minority class (such as abnormal samples) is set to be larger to increase its proportion in the loss calculation.

Prior to model training, it is crucial to normalize the sensor data with different dimensions. This ensures the fairness of the model learning for each feature and prevents the problem of unbalanced gradient updates caused by dimensional differences. The following is a unified dimension-handling solution for the agricultural environmental data of fields and greenhouses:

Min-Max normalization is applicable to data with clear boundaries, such as soil moisture content and relative humidity. It scales the data to an interval [0,1], while preserving the distribution characteristics of the original data. The formula is as follows:

$$X_{\text{Min-Max}} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{17}$$

where X represents a certain sample value in the original data, X_{min} is the minimum value of this feature in the entire dataset, X_{max} is the maximum value of this feature in the entire dataset, and $X_{\text{Min-Max}}$ is the result after normalization.

Z-score standardization is suitable for data that are approximately normally distributed or have no clear boundaries, such as temperature. It transforms the data into a standard normal distribution with a mean of zero and a standard deviation of unity. The formula is as follows:

$$X_{\text{Z-Score}} = \frac{X - \mu}{\sigma} \tag{18}$$

where X is defined as previously described, μ represents the mean value of this feature across the entire dataset, σ represents the standard deviation of this feature across the entire dataset, and $X_{\text{Z-Score}}$ is the result after standardization.

According to the sliding window size determination method described in Section 2.4, the input data of the field and greenhouse were analyzed, and the results are shown in Fig. 7.

The window sizes for the 0–20 cm and 20–40 cm soil moisture in the field were 717 and 973 (1-hour interval), respectively; the window sizes for the air temperature and relative humidity in the greenhouse were 42 and 39 (5-minute interval), respectively.

3.2. Model comparison analysis

To evaluate the advantages of the proposed model (LGCA-Net) in fusing global and local features, we selected random forest (RF), support vector machine (SVM), multi-scale CNN (MCNN), Transformer, temporal convolutional network (TCN), LSTM, and the series-connected TCN-LSTM fusion model for comparison with the proposed model (LGCA-Net). The branch parameters of both the TCN-LSTM fusion model and the proposed model were consistent with those of their respective single-structure counterparts.

The multi-scale convolution module in this study adopts a three-layer structure, with each layer utilizing three convolution kernels of distinct scales. Consequently, the three values of the kernel_size parameter correspond to the sizes of the different convolution kernels, whereas the three values of the filter parameter represent the number of filters in each of the three layers. The detailed parameters of all the models are presented in Table 2.

Table 3 lists the evaluation metrics of the test sets for the eight models under the field and greenhouse scenarios. Both the RF and SVM

Table 2
Parameter configurations of eight models in anomaly detection experiments.

Model	Parameter setting
RF	n_estimators = 100, max_depth = 5
SVM	kernel = rbf, C = 1
MCNN	kernel_size=(3,5,7), filters=(32,16,8)
TCN	kernel_size=(3,5,7), filters=(32,16,8), dilation_rate=(2,4,8)
LSTM	units=(32,16,8), return_sequences = True
Transformer	head_size = 256, num_heads = 4, ff_dim = 4, mlp_units = 128, num_transformer_blocks = 2
TCN-LSTM	TCN: kernel_size=(3,5,7),filters=(32,16,8), dilation_rate=(2,4,8) LSTM: units=(32,16,8), return_sequences = True
LGCA-Net	MCNN: kernel_size=(3,5,7), filters=(32,16,8), Transformer: head_size = 256, num_heads = 4, ff_dim = 4, mlp_units = 128, num_transformer_blocks = 2

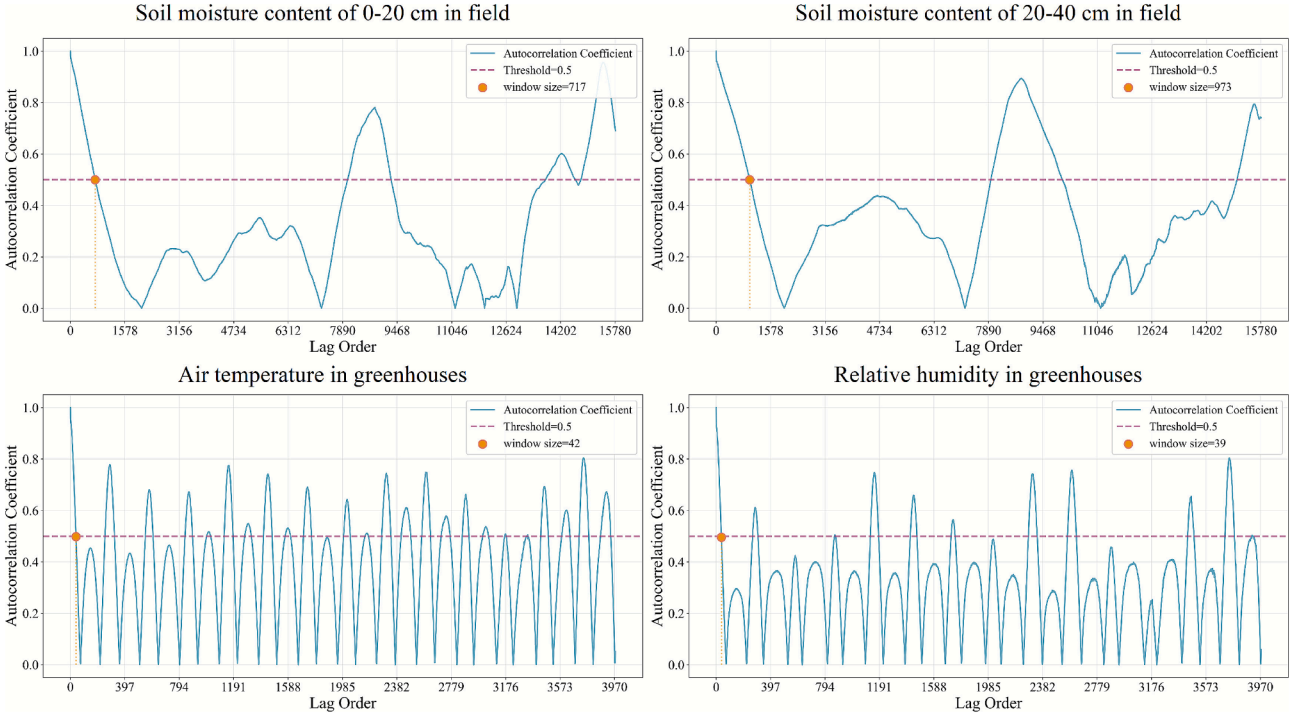


Fig. 7. Autocorrelation coefficient plots of agricultural sensor data in field and greenhouse scenarios.

Table 3
Evaluation metrics of eight anomaly detection models on test sets in field and greenhouse scenarios. sample count: Field (Normal 6577, Abnormal 153); Greenhouse (Normal 2105, Abnormal 12).

Scenario	Model	Recall	Precision	F1-Score	FPR
Field	RF	0.8790	0.5724	0.5910	0.8525
	SVM	0.8601	0.6370	0.6902	0.0468
	MCNN	0.9473	0.9443	0.9452	0.0024
	TCN	0.9215	0.9218	0.9205	0.0035
	LSTM	0.8817	0.9468	0.9115	0.0020
	Transformer	0.9343	0.9645	0.9489	0.0014
	TCN-LSTM	0.9394	0.8787	0.9033	0.0041
	LGCA-Net	**0.9477**	**0.9688**	**0.9580**	**0.0012**
Greenhouse	RF	0.8477	0.7776	0.7901	0.0046
	SVM	0.8992	0.8923	0.8882	0.0017
	MCNN	0.9497	0.9497	0.9497	0.0006
	TCN	0.9495	0.9164	0.9314	0.0011
	LSTM	0.9492	0.8783	0.9078	0.0017
	Transformer	0.9000	0.9994	0.9442	0.0000
	TCN-LSTM	0.9492	0.8747	0.9086	0.0017
	LGCA-Net	**0.9500**	**0.9997**	**0.9721**	**0.0000**

models exhibited significant weaknesses in the two scenarios. In the field (recall 0.8790, precision 0.5724) and greenhouse (recall 0.8477, precision 0.7776) scenarios, the RF suffers from missed detections and false alarms. This limitation results from the reliance of feature engineering on manual design, making it challenging to capture complex correlations in agricultural data. Although the SVM performs marginally better than the RF in the two scenarios, it continues to exhibit limited feature-space mapping capabilities, retaining bottlenecks in missed detections and precision across different scenarios.

Deep learning models, such as MCNN, TCN, LSTM, and Transformers, exhibit advantages in both scenarios because of their automatic feature-learning capabilities. However, their adaptability differs according to the characteristics of the model structures. For the extraction of local features, the MCNN demonstrated excellent recognition accuracy in the field (recall 0.9473, precision 0.9443) and greenhouse (recall 0.9497, precision 0.9497) scenarios. Compared with the TCN model for the data from both scenarios, all MCNN metrics improved, indicating that although the TCN aims to capture multiple scales via dilated convolutions, limitations remain in adapting to complex non-linear relationships and dynamic changes in agricultural time-series data. Specifically, TCN cannot integrate the interactive information between short-term impulses and long-term trends as effectively as MCNN, resulting in incomplete coverage of positive cases and an increase in misjudgments. For the extraction of global features, although the Transformer model, leveraging its self-attention mechanism, has the advantage of uncovering long-term correlations and exhibits significant improvements in most metrics compared with that of LSTM in both scenarios. However, both models exhibit inherent limitations. Although they perform well with long-term soil moisture content data in field environments, both models exhibit reduced adaptability to high-frequency and short-term data in greenhouses. Over-attention to global correlations results in insufficient modeling of key local features. Specifically, compared with the MCNN, the recall metric of the Transformer decreased by 1.3% in the field scenario and 5.0% in the greenhouse scenario.

The TCN-LSTM fusion model, although combining the ability of the TCN to extract multi-scale local features and the ability of the LSTM to capture long-term temporal dependencies, owing to the influence of its tandem structure. The features extracted by the TCN model are used as the input data of the LSTM. If the features extracted by the TCN fail to accurately meet the requirements of the LSTM for long-term temporal features, they can easily cause loss or deviation of the features during the transfer process. In both the field and greenhouse scenarios, this results in the fusion models exhibiting marginally lower or equivalent performance relative to the single-structure models (TCN and LSTM).

The model presented in this paper attains the optimal performance in both the field (recall 0.9477, precision 0.9688) and greenhouse (recall 0.9500, precision 0.9997) scenarios. Compared with models employing single-feature extraction, the proposed model is highly compatible with the characteristics of agricultural time-series data by integrating multi-scale local and global features. For the long-term variations in the soil moisture content in the field, it intensifies the global representation of trends. For the high-frequency periodic correlations of temperature and humidity in the greenhouse, it optimizes the learning of local periodic features. Through the interaction between global and local features, the model can not only capture the long-term change patterns in the field but also precisely respond to the short-term dynamics in the greenhouse. Eventually, it achieved dual control over missed detections and false alarms in both scenarios, with false positive rates (FPR) of 0.0024 and 0.0000, respectively.

Despite its robust performance, the proposed model exhibits potential limitations. The substantially low false positive rate (FPR) of 0.000 observed on the test set of the greenhouse dataset indicates nearly no misjudgment of normal data. However, this result is related to the indicators and quality of the dataset used in this study. The greenhouse dataset in this study included only two indicators—air temperature and relative humidity—with minimal interference from human and natural factors, making the data stable and regular. Additionally, the anomalies in this dataset stem from only a single type: equipment failure. However, in actual agricultural production, the detection indicators for greenhouses are more diverse, covering light intensity, carbon dioxide, and water-fertilizer conditions, among others. Meanwhile, abnormal situations are also more complex. When confronted with these more complex scenarios and anomaly types, the model may still experience missed detections or false alarms.

See Figs. 8 and 9 present the normalized confusion matrices of the eight models evaluated under the field and greenhouse scenarios, respectively. As shows in the figures, in the field scenario, the soil moisture content is influenced by multiple factors such as irrigation, precipitation, and seasonal cycles, making the boundary between “normal” and “abnormal” ambiguous. For example, short-term high soil moisture content caused by irrigation is a normal fluctuation rather than an abnormality. For traditional models such as RF and SVM, the distribution of the confusion matrix along the diagonal is uneven. The recognition rate of the RF for normal samples was only 67.72%, indicating that a substantial number of normal soil moisture fluctuations were misclassified as abnormal. RF cannot distinguish between “natural fluctuations” and “true abnormalities.” The recognition rate for the abnormal samples was 88.45%. Although it can capture extreme abnormalities such as sudden changes caused by sensor failures, the false alarm rate is high, reflecting a limited understanding of agricultural time-series characteristics. The recognition rate of the SVM for normal samples increased to 95.32%; however, the recognition rate for abnormal samples was only 77.12%. Similar to RF, SVM fails to capture complex abnormal patterns under the combined effects of precipitation and irrigation, resulting in missed detection of most abnormalities.

Among deep-learning models such as MCNN, TCN, LSTM, Transformer, and TCN-LSTM, the recognition accuracy of normal samples significantly increased compared with traditional machine learning models, reaching more than 99.50% in all cases. However, performance differences emerge in abnormal sample recognition. The MCNN, owing to its multiscale convolutions, effectively captures multidimensional differences in soil data, achieving abnormal recognition rate that exceeds that of TCN by 3.27%. The self-attention mechanism of a Transformer can focus dynamically on key abnormal features. Compared with the sequential calculation of LSTM, it exhibits a higher recognition accuracy for sparse abnormalities (abnormal events with low frequency and high impact), and its abnormal recognition rate increases by 5.88%. Although the TCN-LSTM fusion model improved the recognition rate of abnormal samples compared with the single-structured TCN and LSTM, owing to the influence of serial feature transmission, it also resulted in

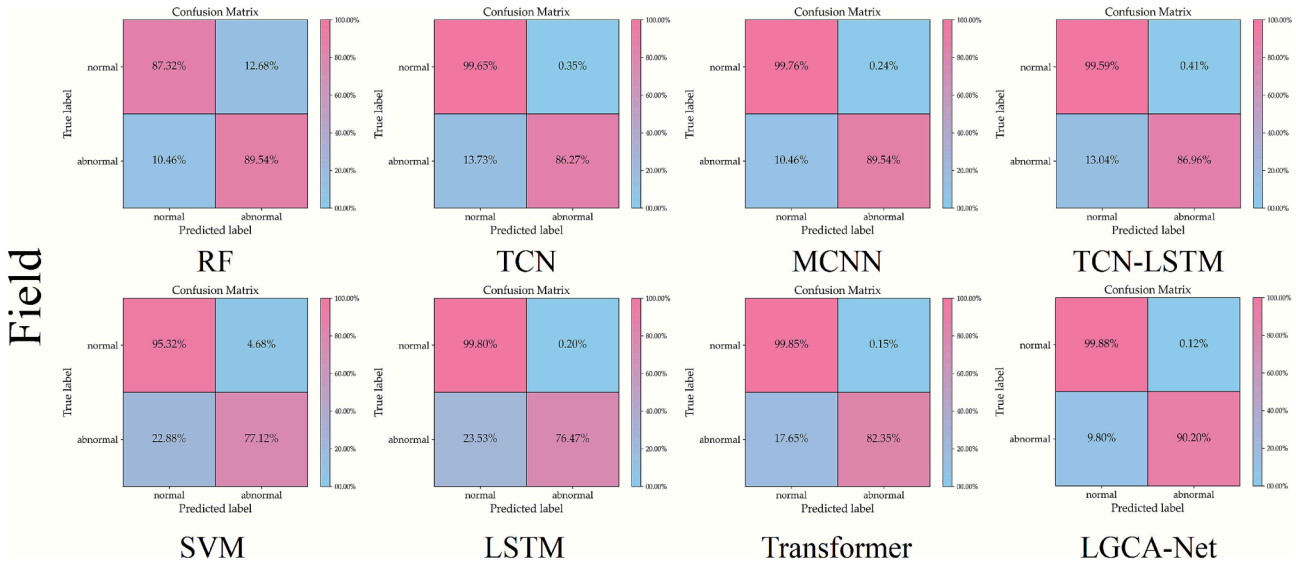


Fig. 8. Normalized confusion matrices of eight anomaly detection models on the field scenario test set.

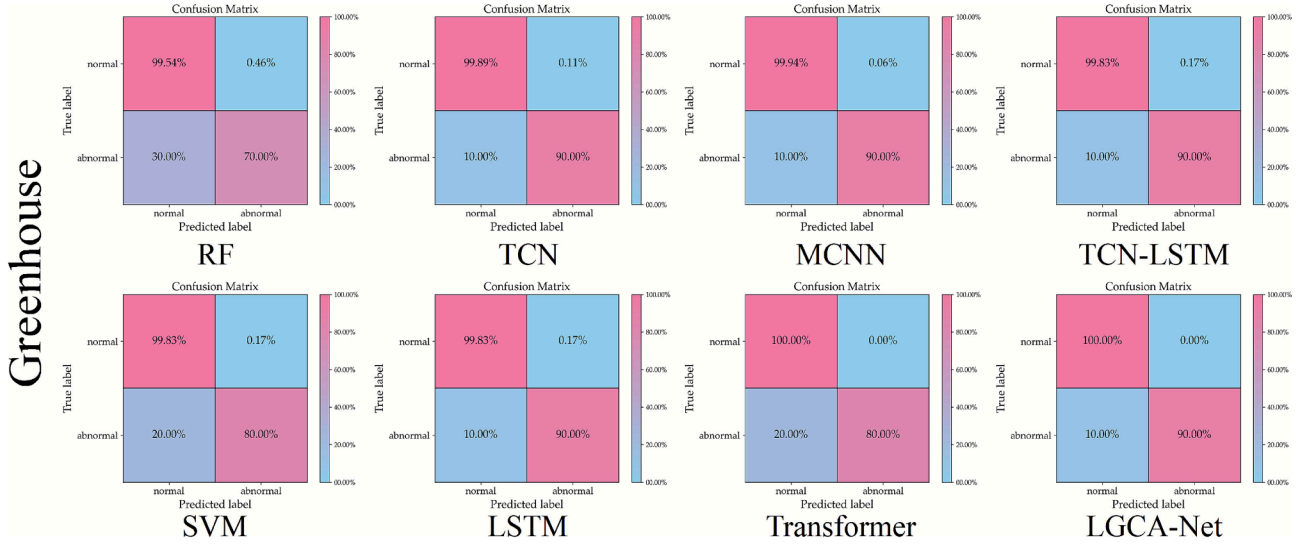


Fig. 9. Normalized confusion matrices of eight anomaly detection models on the greenhouse scenario test set.

an increase in the false detection rate of normal samples. The model proposed in this paper, through the fusion of MCNN and Transformer, achieved a recognition accuracy of both normal and abnormal samples exceeding 90%, which is higher than that of other models. Thus, it can be seen that the effectiveness of each model in detecting anomalies largely depends on how well its architecture covers different anomaly patterns.

Compared with the field scenario, results obtained from the greenhouse dataset remain largely consistent, except that the LSTM model exhibits an abnormal recognition accuracy approximately 10% higher than that of the Transformer. This is likely attributed to the influence of data sampling frequency. Greenhouse data are collected at higher temporal resolution, yielding denser time-series information characterized by frequent short-term fluctuations and instantaneous features. The Transformer, which emphasizes global features and is likely to "overlook" these short-term local abnormalities, may misjudge the abnormalities in high-frequency normal fluctuations as normal, thereby limiting the abnormal recognition accuracy. However, by combining the local features extracted by the MCNN model, the Transformer abnormal recognition accuracy is improved while ensuring that the accuracy of normal sample recognition does not decline. It can be seen that for the fusion model proposed in this paper, both low-frequency field monitoring scenarios that emphasize long-term trends and high-frequency greenhouse scenarios that require detection instantaneous fluctuations and subtle abnormalities benefit from the coordinated global and local feature extraction. Global feature extraction captures long-range correlations and overarching temporal patterns, whereas local features extract short-term and fine-grained abnormal patterns. Their effective interaction enables the model to accurately characterize normal samples while sensitively identifying abnormal patterns, demonstrating strong robustness across heterogeneous agricultural monitoring conditions.

3.3. Cross-Attention mechanism ablation analysis

To evaluate the advantages of the proposed cross-attention feature fusion, it was compared with the traditional feature concatenation. The experiments were performed under both field and greenhouse scenarios. The results are illustrated in Fig. 10.

As shown in Fig. 10(a), for the field data, the feature fusion method using the cross-attention mechanism exhibited an overall improvement

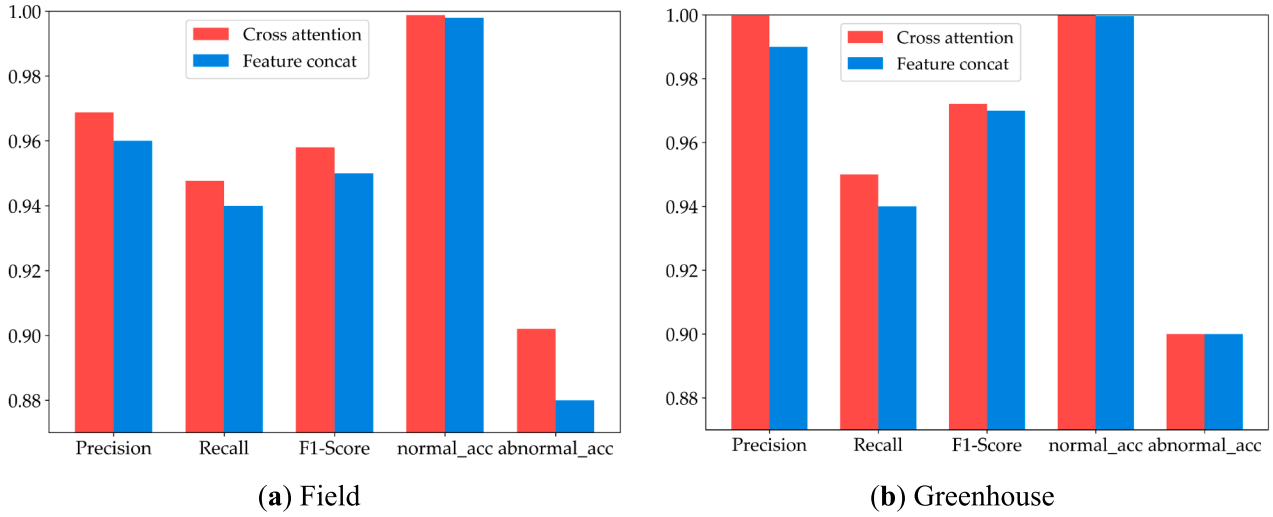


Fig. 10. Ablation experiment comparison of cross-attention in field and greenhouse scenarios.

in the performance indicators. In particular, the recognition accuracy of the abnormal samples showed the most significant increase. This is likely attributed to the greater influence on field soil data by the natural environment. Simple feature concatenation is prone to noise interference, whereas cross-attention dynamically emphasizes key features and suppresses irrelevant noise, thereby improving the recognition of abnormal samples.

Compared with the field soil data, the greenhouse environmental data were less disturbed. Therefore, as shown in Fig. 10(b), although the cross-attention mechanism marginally improves the performance in terms of precision and recall, the detection accuracy of the normal and abnormal samples is close to that of the traditional feature concatenation method. These results indicate that the benefits of cross attention are more pronounced for complex datasets.

Finally, to verify the practical effect of the proposed model in detecting abnormal data outside the training samples, for the four elements in the field and greenhouse scenarios, a continuous time series outside the training sample time was used for real-time abnormal detection. The results are shown in Figs. 11–14 (where the blue dots represent samples with incorrect abnormal recognition, and the red crosses represent samples with correct abnormal recognition).

The figures indicate suboptimal performance of traditional models (RF and SVM) in detecting soil moisture content, temperature, and humidity. This indicates that they are only suitable for scenarios with extremely simple features and single abnormal patterns. The TCN and MCNN are effective at handling scenarios with significant local features and numerous minor fluctuations. For the 0–20 cm shallow soil moisture content dataset, their anomaly detection accuracy outperforms that of LSTM and Transformer. By contrast, the LSTM and Transformer exhibited stable performance in scenarios with periodic data (such as greenhouse humidity) and prominent anomalies (such as deep soil moisture content). Their ability to learn long-sequence dependencies is compatible with the characteristics of time-series data. However, in terms of identifying local minor fluctuations, they were marginally inferior to TCN and MCNN. Although the TCN-LSTM fusion model aims to combine local feature extraction with long-sequence modelling capabilities, its performance is highly dependent on the quality of feature extraction from the TCN front end because of the limitations of its serial structure, resulting in only marginal improvements.

The proposed LGCA-Net model exhibits lower overall misjudgment rate than that of the traditional models, and demonstrates optimal robustness in different scenarios. In the detection of the 0–20 cm shallow field soil moisture content, this model exhibited a relatively high anomaly misjudgment rate, with misjudgments primarily concentrated in the segment where the soil moisture content at the end of the data had repeated minor changes. This is because shallow soils are more sensitive to human and environmental disturbances. Simultaneously, the mutation range of the end-part data is small, which increases the difficulty of identification. In the identification of the 20–40 cm deep field soil moisture content, although frequent mutations were present at the end owing to the larger data change amplitude, the model was capable of effectively detecting abnormalities. In the greenhouse scenario, the detection of the environmental temperature and relative humidity was satisfactory. Except for two minor abnormal fluctuations in the relative humidity that were not identified, nearly all other abnormal samples were accurately detected.

In summary, the proposed abnormal detection model of global and local feature fusion exhibits robust detection capabilities for sensor data in different agricultural environments. Although there is room for optimization owing to the influence of data features in complex scenarios such as shallow soil, it performs reliably in deep soil and regular greenhouse environments, providing valuable technical support for agricultural environment monitoring. In the future, the model structure can be further optimized for special scenarios to improve the model's ability to identify minor fluctuating abnormalities.

3.4. Analysis of model complexity and inference efficiency

To further evaluate the applicability of the proposed LGCA-Net model to real-time agricultural monitoring scenarios, we statistically analyzed and compared the parameter scale, model size, computational complexity (FLOPs), and average inference time of various models. Because traditional models (RF and SVM) exhibit low accuracy and cannot leverage the GPU for inference acceleration, this section focuses on a horizontal comparison among deep learning models (TCN, MCNN, LSTM, Transformer, TCN-LSTM). To ensure a fair comparison, all the experiments were conducted under the same hardware environment (CPU: Intel Core i7-14700F; memory: 32 GB; GPU: NVIDIA RTX 4060 Ti (8G)). The results are summarized in Table 4.

As shown in Tables 3 and 4, the LSTM and TCN-LSTM models are limited by the LSTM network structure and cannot achieve parallel computing. Their inference times ranged from 42.296 ms to 50.881 ms, resulting in suboptimal real-time performance and lower accuracy than that of LGCA-Net. They only have lightweight advantages in terms of computational complexity (12.12 M−34.26 M) and model size (86 KB–167 KB), making them suitable only for simple scenarios with limited computing performance and low requirements for both real-time performance and accuracy.

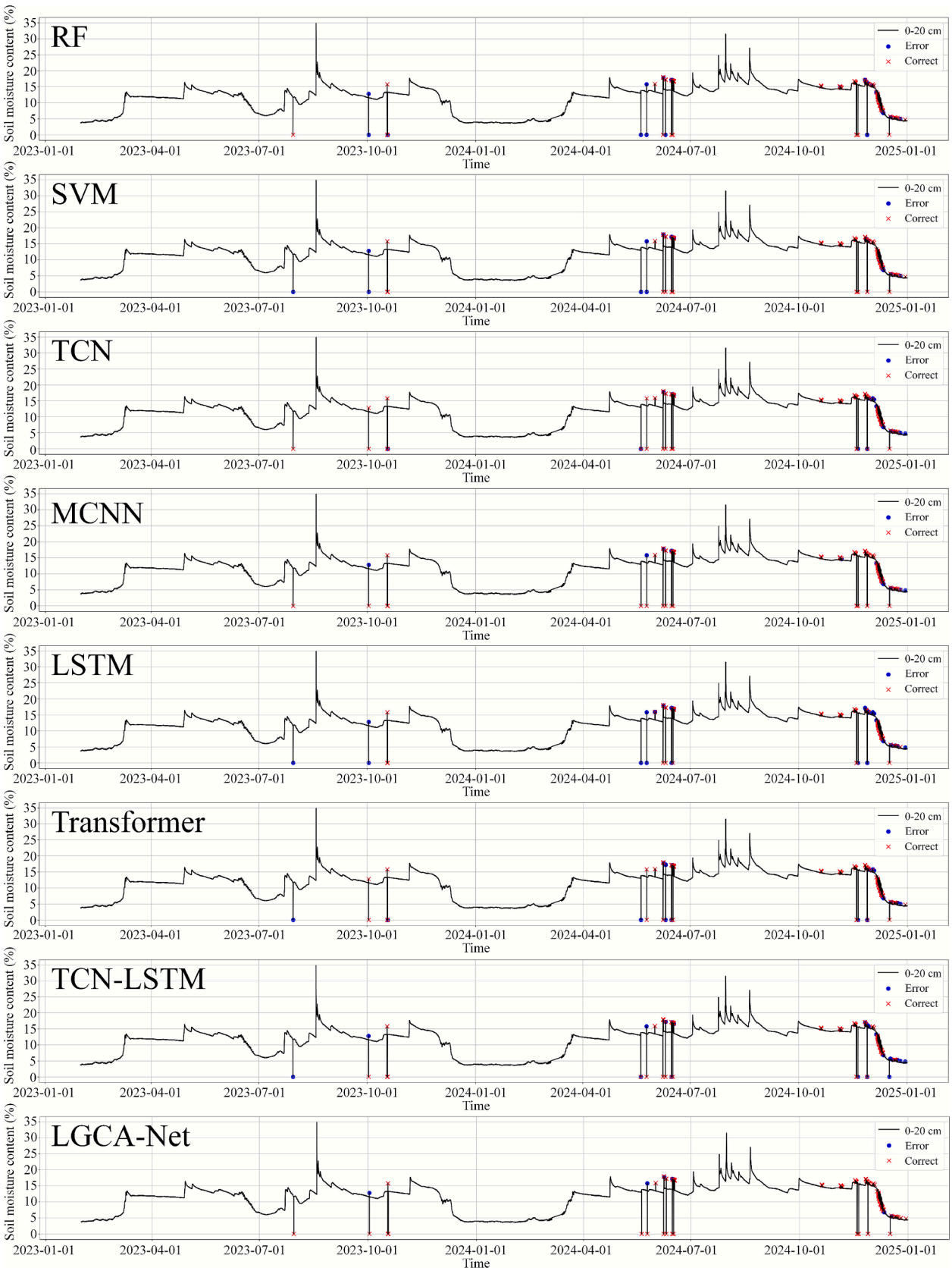


Fig. 11. Detection results of abnormal soil moisture content of 0–20 cm in field.

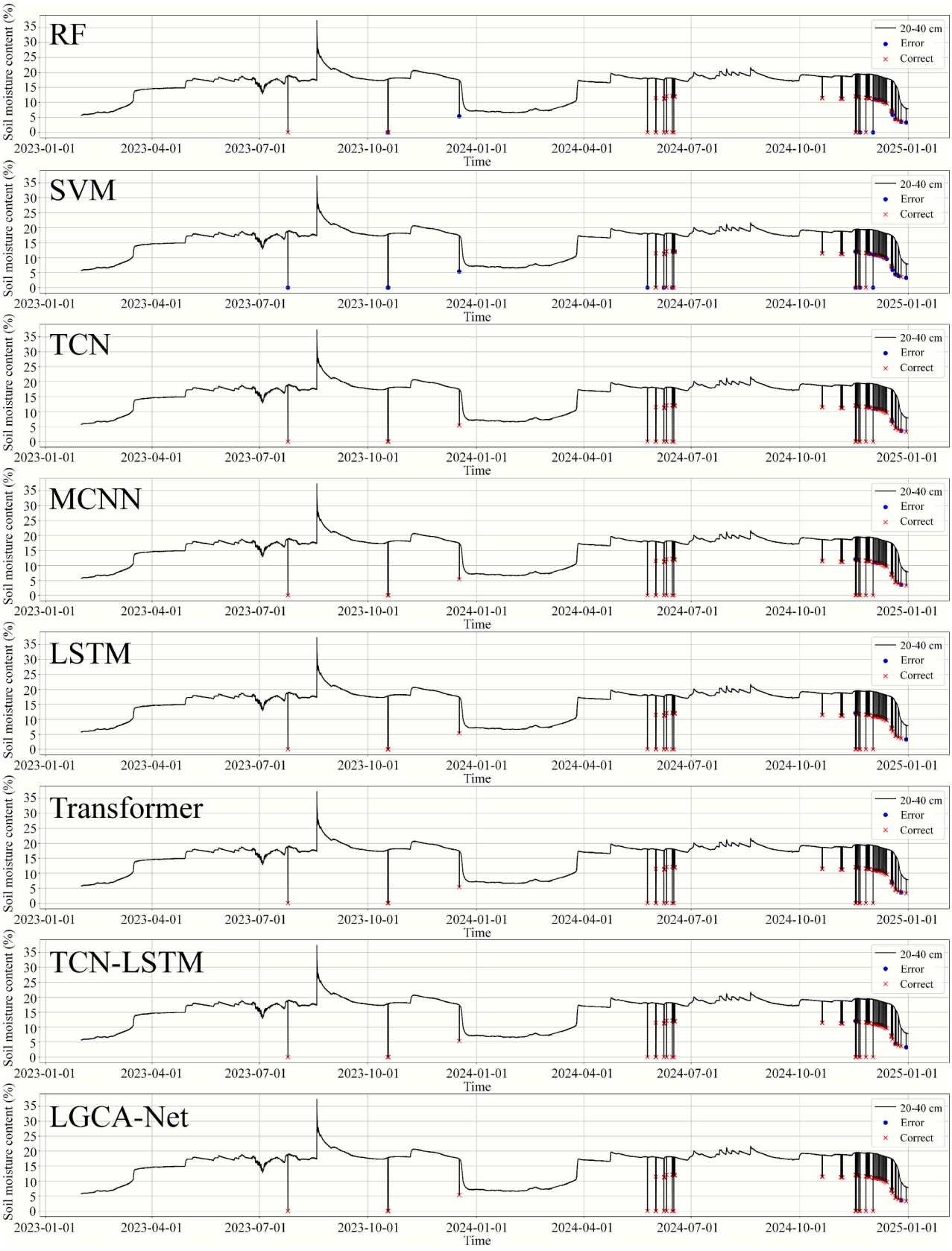


Fig. 12. Detection results of abnormal soil moisture content of 20–40 cm in field.

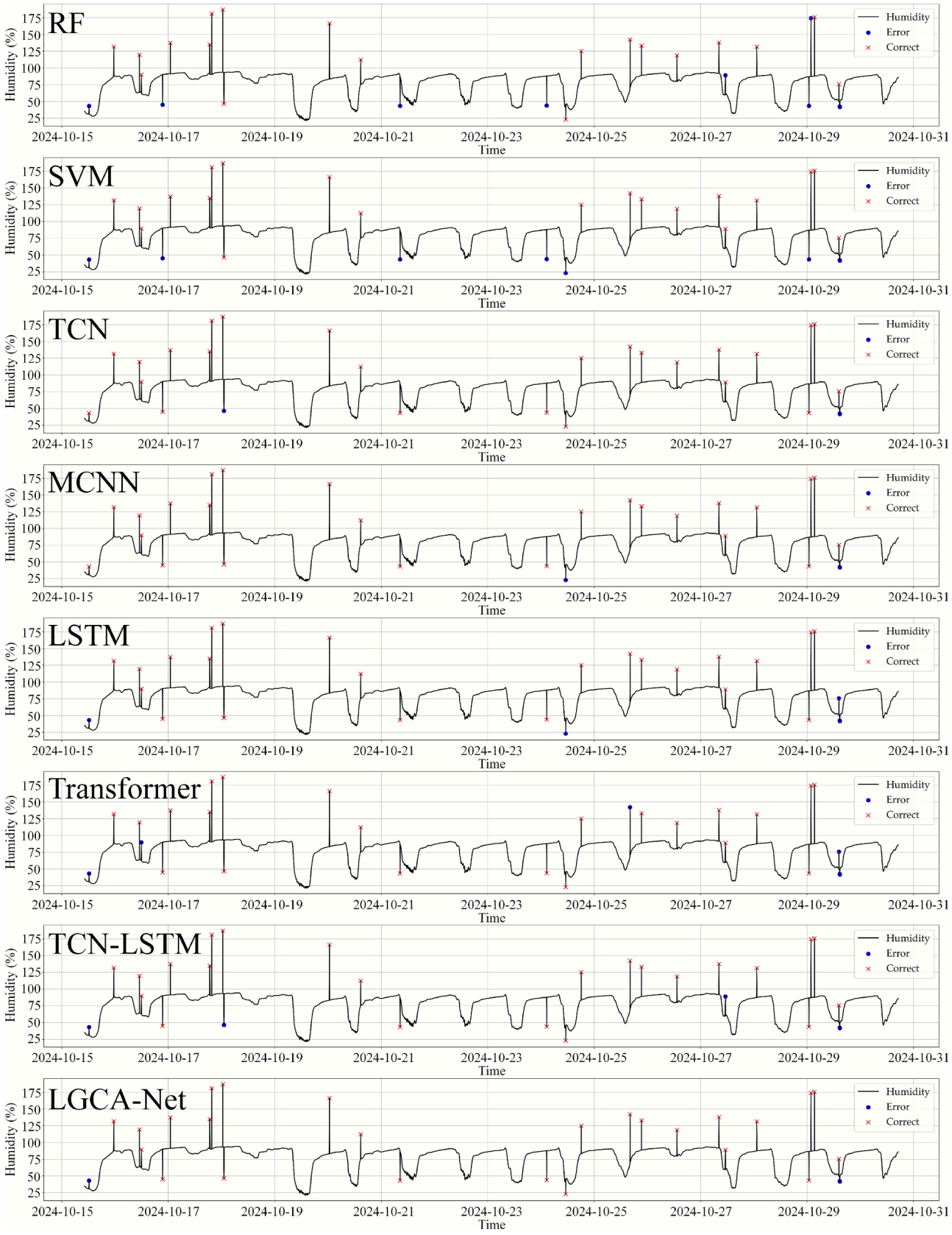


Fig. 13. Detection results of abnormal air relative humidity in greenhouse.

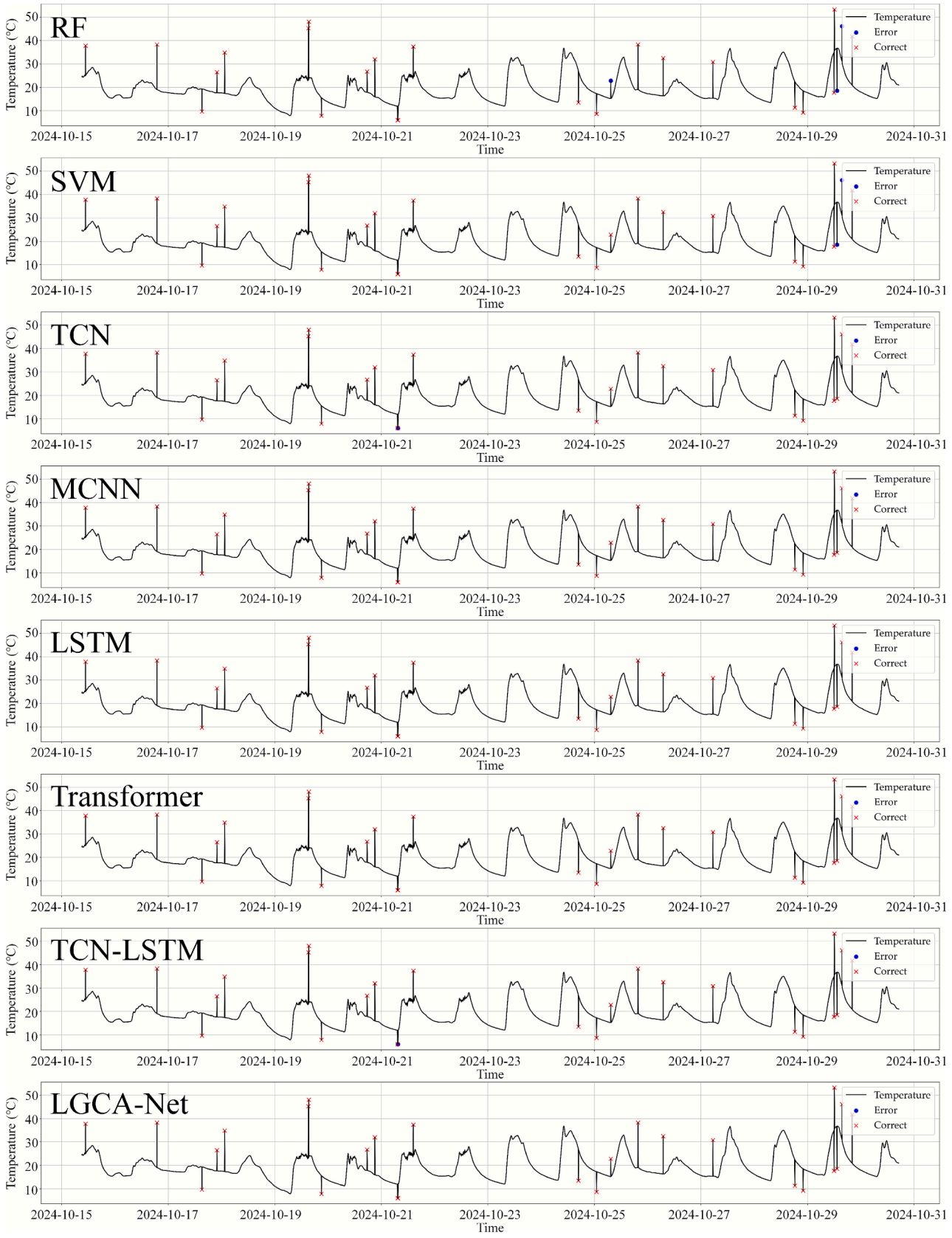


Fig. 14. Detection results of abnormal air temperature in greenhouse.

Table 4
Comparison of parameter scale, model sizes, computational complexity (FLOPs), and inference performance of six deep learning models for agricultural sensor time-series anomaly detection.

Model	Parameter Scale (units)	Model Size (KB)	FLOPs (M)	Average Inference Time (ms)
MCNN	40,849	266	73.13	14.124
TCN	19,281	118	21.65	10.518
LSTM	15,497	86	12.12	42.296
Transformer	14,757	120	34.41	10.500
TCN-LSTM	28,465	167	34.26	50.881
LGCA-Net	55,605	388	107.55	20.209

Although TCN and Transformer are lightweight (size: 118–120 KB) and have fast inference speed (approximately 10.5 ms), their accuracy is insufficient, making it difficult to balance "speed and performance". Although MCNN and the proposed LGCA-Net have higher parameters, larger sizes, and greater computational complexity than the aforementioned models, the proposed LGCA-Net achieves a more significant improvement in accuracy. Moreover, its inference time of 20.21 ms is substantially lower than the sampling frequencies of 5 min and 1 h, meeting the requirements of real-time monitoring and delivering higher comprehensive cost-effectiveness under limited edge-side resources. Additionally, the proposed model extracts more comprehensive features, and its accuracy advantage will be more prominent in complex scenarios.

4. Conclusion

In response to the long-term and short- erm variation patterns exhibited by agricultural time-series sensor data, this study introduced a general time-series anomaly detection method for multiple types of agricultural sensors, LGCA-Net, based on the fusion of global and local features. This method leverages a multi-scale CNN (MCNN) to extract local features, captures global features via the multi-head attention mechanism of the transformer, and implement the interactive integration of global and local features employing the cross-attention mechanism. Consequently, anomaly detection can be effectively performed for time-series data in diverse agricultural scenarios.

The experimental results indicate that, compared with the seven existing models, the proposed fusion model demonstrates superior detection performance in both field and greenhouse scenarios. In terms of the F1-Score metric, which comprehensively measures the balance between false alarms (precision) and missed detections (recall), the model achieved 0.9580 in the field scenario and 0.9721 in the greenhouse scenario. Compared with the single-structure models, it exhibited an average increase of 6.65% (in the field) and 6.67% (in the greenhouse) in the recognition accuracy of abnormal samples.

In addition, to evaluate the advantages of the cross-attention fusion mechanism, we conducted an ablation analysis. The results show that the optimization effect of cross attention is particularly significant when the data are affected by complex factors. When the influencing factors of the data are simple, their effect is similar to or marginally better than that of the traditional concatenation method. This indicates that the proposed model is more suitable for handling complex agricultural data. Meanwhile, LGCA-Net maintains good computational efficiency, with an average inference time of 20.209 ms under the experimental hardware environment, which is substantially lower than the sampling frequencies of the field (1 h) and greenhouse (5 min) scenarios, meeting real-time monitoring requirements.

Although this study has achieved significant progress, several limitations remain that can be further addressed in future work. (1) The experimental validation was limited to datasets from two scenarios, field and greenhouse, which constrain the full demonstration of the model's performance. In the future, we will expand the research scope to include agricultural sensor data from more types and application scenarios to further explore the generalizability of the model in complex environments; (2) This study employs a supervised learning framework, where the range of detectable anomalies is constrained by the diversity of annotated anomaly samples in the dataset. Therefore, practical deployment requires fine-tuning for specific datasets to adapt to different anomaly situations. Future research will further optimize the model structure and attempt to introduce self-supervised learning methods, enabling the model to independently learn the distribution characteristics of normal data, mitigate the reliance on annotated samples; (3) We will also take into account the requirements of edge-end deployment and focus on the lightweight optimization of the model to enhance its applicability in resource-constrained agricultural IoT scenarios.

CRediT authorship contribution statement

Zongren Wang: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Laixi Zhang:** Supervision. **Jianbo Guo:** Supervision. **Yuan Huang:** Investigation. **Ping Lin:** Project administration. **Wengang Zheng:** Supervision, Funding acquisition. **Lili Zhangzhong:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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